

ePortfolio Update 2 - Reflection Essay

Oliver Braitsch

Introduction

For this reflection draft, I am connecting my capstone experience on Team C603, the AIAA Oregon State Student Competition Rocket Team roll-control project, to the NACE career readiness competencies. Our project focused on designing an active roll-control system for a solid fuel competition rocket using servodriven trim tabs. At the start of the year, the roll-control part of the project was very open-ended. We did not have a finished prototype, completed trade studies, or a proven system, so a lot of the work involved turning a broad goal into a design that could actually fit in the rocket and be tested.

The competencies below are the ones I think connect most directly to my individual experience. For the final version, I may polish the wording and add more final results from the expo/prototype, but this draft shows the main ways capstone has prepared me for engineering work after graduation.

Critical Thinking

A major part of this project that addressed critical thinking was narrowing the roll-control concept into a trim tab system that could fit in the fin and fairing while still creating enough roll authority. The project was not just one calculation or one CAD model. I had to think through competing constraints such as control authority, drag, competition safety rules, servo packaging, stiffness, and assembly complexity.

One specific example was deciding how the trim tab, hinge shaft, clamp, and servo connection could work together inside a very thin fin. The tab had to rotate, but the mechanism also had to stay nearly flush so it would not create extra aerodynamic problems. This made me compare different design options and think about whether each one would actually be buildable, not just whether it looked good in CAD.

This prepares me for a career after graduation because aerospace and mechanical engineering problems are usually constrained and imperfect. I learned that I need to use analysis to support decisions, but I also need to think practically about manufacturing, space, assembly, and system-level requirements.

Technology

The technology competency was addressed through my use of Onshape and other engineering tools to turn ideas into a usable design. I used CAD throughout the project to model the trim tab, hinge shaft, clamp, coupler, servo interface, and housing components. The CAD work was not only for presentation; it helped reveal whether the design could physically fit and move correctly.

A concrete example was checking the trim tab assembly inside the 0.125-inch-thick fin. I used the model to look for interference, understand how the servo motion would transfer to the tab, and adjust parts when the first layout was too tight or created assembly issues. I also had to work through practical CAD issues like mates, rotation behavior, and how different parts should be connected in the assembly.

This experience prepares me for professional engineering because modern design work depends heavily on digital tools. I became more comfortable using CAD as a problem solving tool, not just a way

to make parts look complete. In a future aerospace role, that will help me communicate designs, catch problems earlier, and revise concepts quickly when requirements change.

Communication

Communication was important because the roll-control system touched several parts of the rocket project. My mechanical design choices affected the fincan, fairing, servo placement, battery and protoboard housing, screw locations, and how the prototype could be assembled. Because of that, I had to explain design changes and ask questions when there were conflicts between parts. One example was communicating interface needs for the trim tab mechanism and the internal housings. When I changed mounting tabs, hole sizes, or the servo-to-tab connection, those changes mattered to teammates who were printing parts, assembling the fincan, or planning the electronics layout. I had to be clear about what changed and why, especially when the design was changing quickly near the end of the Term.

This prepares me for work after graduation because engineering is not only about doing technical work by myself. A design has to be understood by other people before it can be reviewed, manufactured, tested, or improved. Capstone gave me practice communicating technical details in a team environment where unclear information could slow the whole project down.

Teamwork

Teamwork showed up throughout the project because each person had individual tasks, but the final system depended on all of the parts working together. My work on the trim tab and mechanical integration had to line up with teammates working on printing, electronics, the fincan, and other parts of the rocket.

A specific example was when the quick prototype connection using thin wire created too much wobble. That issue affected more than just one part; it changed how reliable the actuation looked and whether the prototype would be convincing for a demonstration. The team had to think through a practical fix using available hardware, stiffness, and assembly time. I also helped keep track of parts and adjusted CAD designs as constraints changed.

This prepares me for an engineering career because real projects depend on shared interfaces and shared responsibility. I learned that being a good teammate means more than finishing my assigned section. It also means helping solve problems when they affect the larger system and being flexible when another person's work changes my design.

Leadership

Leadership in this project was less about having a formal title and more about taking ownership of pieces of the design that needed to move forward. There were many small mechanical details that had to be solved before the system could work, including the servo and tab connection, housing attachment method, screw selection, and final assembly approach. One concrete example was taking initiative on the physical integration details instead of waiting for the design to be perfectly defined. I worked through options, made reasonable design choices, brought those choices back to the team, and revised them when testing or assembly showed a better solution was needed. This was important because the project started with no working prototype and needed to become something physical by the end of capstone. This

prepares me for a career because entry-level engineers still need to take ownership of their work. I learned that leadership can mean keeping progress moving, identifying what still needs to be solved, and helping the team stay focused on the final goal even when the details are uncertain.

Professionalism

Professionalism was addressed through the need to complete useful work on a real deadline. As the project moved from CAD into physical parts, small details became much more important. Screw size, hole clearance, printed part tolerances, material stiffness, and assembly access all affected whether the design would actually work. A specific example was the end-of-project push to assemble and improve the prototype for capstone deliverables and the expo demonstration. I had to be dependable in tracking parts, picking up components, updating CAD, and making sure my portion of the system was ready for integration. This made the project feel more like professional engineering work than a normal class assignment. This prepares me for work after graduation because professional engineers are judged by the reliability and accuracy of their work, not just by the original idea. Capstone helped me see that professionalism includes being prepared, paying attention to details, communicating delays or problems, and following through on the parts of the project I am responsible for.

Career & Self-Development

Career and self-development connects to this project because it helped confirm the type of engineering work I want to pursue. I am interested in aerospace engineering, especially systems where mechanical design affects flight performance. The roll-control project connected CAD, mechanisms, aerodynamics, controls, and prototyping in a way that felt very relevant to my career goals. One concrete lesson was realizing that a design can look clean in CAD but still need major changes when real parts, tolerances, fasteners, and assembly access are involved. That showed me areas where I still want to improve, especially hardware testing, controls integration, and compact mechanism design. It also gave me more confidence in the skills I have developed through coursework and project work. This prepares me for my career because it gave me a clearer understanding of my strengths and areas for growth. After graduation, I want to keep building skills in CAD, analysis, testing, and aerospace system design so I can contribute to engineering teams working on flight hardware or related mechanical systems.

Conclusion

Overall, capstone has helped me connect classroom learning to the type of work I expect to do as a mechanical/aerospace engineer. The SCRT roll-control project required critical thinking, technology, communication, teamwork, leadership, professionalism, and career self development. The biggest takeaway for me is that these competencies overlap in real engineering work. A design decision also requires communication, teamwork, documentation, and practical judgment. For the final reflection, I plan to polish this draft and possibly add more detail from the final prototype/demo, but this version captures the main ways the project has prepared me for life after graduation.